Introduction to Quantitative Biology
Lecture Notes: The Origin and Propagation of Noise in Gene Expression

From Biological Circuit Design — Michael Elowitz and Justin Bois

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This lecture note is based primarily on Michael Elowitz and Justin Bois's course on "*Biological Circuit Design*, and Johan Paulsson's 2004 Nature paper,"*Summing up the noise in gene networks*," together with its Supplementary Information (SI). Our derivation also draws on the Linear Noise Approximation approach developed by Johan Elf and Måns Ehrenberg in "*Fast Evaluation of Fluctuations in Biochemical Networks With the Linear Noise Approximation*."

1. Noise in Gene Expression

Random fluctuations in genetic networks are inevitable as chemical reactions are probabilistic, and many genes, RNAs, and proteins occur in small numbers per cell. Roughly speaking, we can call random deviations from what we might expect in a deterministic view of gene expression **stochasticity**, or **noise**.

1.1 Intrinsic and Extrinsic Noise

All cell components display **intrinsic noise** due to random births and deaths of individual molecules, and **extrinsic noise** due to fluctuations in reaction rates. The terms 'intrinsic' and 'extrinsic' generically distinguish between the origin and propagation of noise, and their biological meaning is always defined in conjunction with a specified component or process.

For example, if a gene for a transcriptional repressor spontaneously switches on and off, thereby enslaving the encoded protein and transmitting the fluctuations to repressed genes, the noise is intrinsic to the number of active repressor genes and extrinsic to all affected components.

1.2 Measures of Gene-Expression Noise

For a molecular count n_i with mean $\langle n_i \rangle$, we use σ_i to denote the standard deviation, and σ to denote the covariance matrix:

$$\sigma_i^2 = \langle (n_i - \langle n_i \rangle)^2 \rangle, \qquad \sigma_{ij} = \langle (n_i - \langle n_i \rangle)(n_j - \langle n_j \rangle) \rangle, \qquad \sigma_{ii} = \sigma_i^2.$$

Paulsson compares the relative variance and the Fano factor:

$$\text{Relative variance} = \frac{\sigma_i^2}{\langle n_i \rangle^2}$$

$$\text{Fano factor} = \frac{\sigma_i^2}{\langle n_i \rangle}$$

The two measures therefore differ by a factor of $\langle n_i \rangle$. For a Poisson distribution, the Fano factor equals one. Paulsson uses relative variance to quantify gene-expression noise in Eq. (1) of his 2004 paper. We will derive this equation and explain the choice of noise measure in the next lecture note.

2. Single-Species Birth–Death Process and Master Equation

Chemical reactions in single cells in general occur far from thermodynamic equilibrium, and the molecule copy numbers can sometimes be very small. Both these properties make it mandatory to analyze the stochastic behavior of chemical reactions in living systems. One technique is to solve the **chemical Master equation**, which describes the time evolution of the probability distribution for all the different molecular components in the system.

We first consider the simplest single-species birth–death process,

$$\emptyset \xrightarrow{birth} X \xrightarrow{death} \emptyset$$

$$\emptyset \xrightarrow{k} X, \qquad X \xrightarrow{\gamma} \emptyset$$

Here k is the production rate, while each molecule is degraded independently at rate γ. If there are n molecules of X, the total degradation rate is therefore γn.

At the macroscopic level, the molecule number satisfies

$$\frac{dn}{dt} = k - \gamma n.$$

The deterministic stationary copy number is

$$n^* = \frac{k}{\gamma}.$$

To describe the stochastic dynamics, let $P_n(t)$ denote the probability of having n molecules at time t. The possible transitions are

$$n-1 \xleftarrow{\gamma n} n \xrightarrow{k} n+1.$$

The corresponding Master equation is

$$\frac{dP_n}{dt} = kP_{n-1} + \gamma(n+1)P_{n+1} - kP_n - \gamma n P_n,$$

The first two terms are probability inflows to state n, while the last two terms are probability outflows from state n.

2.1 Stationary Solution of the Master Equation

We now look for the stationary probability distribution, for which

$$\frac{dP_n}{dt} = 0.$$

The birth–death process can be represented as transitions between neighboring molecule-number states,

$$P_0 \underset{\gamma}{\overset{k}{\rightleftharpoons}} P_1 \underset{2\gamma}{\overset{k}{\rightleftharpoons}} P_2 \underset{3\gamma}{\overset{k}{\rightleftharpoons}} \cdots$$

The forward transition from state n to state $n+1$ occurs with rate k, whereas the reverse transition occurs with rate $\gamma(n+1)$.

We define the net probability flux from state n to state $n+1$ as

$$J_n = kP_n - \gamma(n+1)P_{n+1}.$$

The Master equation can then be written in terms of the probability fluxes as

$$\frac{dP_n}{dt} = kP_{n-1} + \gamma(n+1)P_{n+1} - kP_n - \gamma nP_n = J_{n-1} - J_n.$$

At stationarity, we have

$$J_{n-1} = J_n.$$

Thus the **probability flux** is the same between every pair of neighboring states,

$$J_0 = J_1 = J_2 = \cdots$$

For a normalizable stationary distribution, there can be no finite probability current flowing to arbitrarily large molecule numbers. Hence, in the limit $n \to \infty$,

$$J_n \to 0.$$

Since the stationary flux is independent of n, this implies

$$J_n = 0 \qquad \text{for all } n.$$

Therefore,

$$kP_n = \gamma(n+1)P_{n+1} \tag{2.1}$$

This is the **detailed balance condition**: at stationarity, the probability flux from n to $n+1$ is exactly balanced by the reverse flux from $n+1$ to n.

Equation (2.1) gives

$$P_{n+1} = \frac{k}{\gamma(n+1)}P_n.$$

Applying this relation successively from state 0 to state n gives

$$\begin{aligned} P_n &= P_0 \prod_{m=1}^{n} \frac{k}{\gamma m} \\ &= P_0 \frac{1}{n!} \left(\frac{k}{\gamma}\right)^n . \end{aligned} \tag{2.2}$$

The probability distribution must satisfy

$$\sum_{n=0}^{\infty} P_n = 1.$$

Using Eq. (2.2),

$$\begin{aligned} 1 &= P_0 \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{k}{\gamma}\right)^n \\ &= P_0 \exp\left(\frac{k}{\gamma}\right) \end{aligned}$$

Therefore,

$$P_0 = \exp\left(-\frac{k}{\gamma}\right),$$

and the stationary probability distribution follows a **Poisson distribution**,

$$P_n = \exp\left(-\frac{k}{\gamma}\right) \frac{(k/\gamma)^n}{n!} \sim \text{Poisson}\left(\frac{k}{\gamma}\right).$$

Its mean and variance are

$$\langle n \rangle = \frac{k}{\gamma}, \qquad \sigma^2 = \frac{k}{\gamma}.$$

The **coefficient of variation** and **Fano factor** are therefore

$$\text{CV} = \frac{\sigma}{\langle n \rangle} = \frac{1}{\sqrt{\langle n \rangle}},$$

$$F = \frac{\sigma^2}{\langle n \rangle} = 1.$$

Thus, although the absolute variance increases with expression level, the relative fluctuations decrease as the average molecule number increases.

3. Linear Noise Approximation

The Master equation can be written directly from the rate constants and stoichiometries of all the elementary reactions, but analytical and numerical solutions are generally difficult to obtain. It can often be simplified through van Kampen's Ω-expansion, which gives a **Linear Noise Approximation** (LNA) near macroscopic trajectories or stationary points.

3.1 Macroscopic and Mesoscopic Dynamics

We now consider a single chemical species X in a system of volume Ω. The molecule number is denoted by n, and the macroscopic concentration by x.

Consider a general single-species birth–death process,

$$\emptyset \xrightarrow{R_+(n)} X, \qquad X \xrightarrow{R_-(n)} \emptyset.$$

The possible transitions are

$$n-1 \xleftarrow{R_-(n)} n \xrightarrow{R_+(n)} n+1.$$

A birth event changes the molecule number by $+1$, while a death event changes it by -1.

In the macroscopic limit,

$$x = \lim_{\Omega\to\infty} \frac{n}{\Omega},$$

and the transition rates per unit volume approach the macroscopic reaction rates

$$r_\pm(x) = \lim_{\substack{\Omega\to\infty \\ n/\Omega\to x}} \frac{R_\pm(n;\Omega)}{\Omega}.$$

The corresponding Master equation for the molecule-number probability $P_n(t)$ is

$$\frac{\partial P_n}{\partial t} = R_+(n-1)P_{n-1} + R_-(n+1)P_{n+1} - [R_+(n) + R_-(n)]\,P_n \tag{3.1}$$

Define the shift operators

$$E^\pm f(n) = f(n \pm 1).$$

Equation (3.1) can then be written as

$$\frac{\partial P}{\partial t} = (E^- - 1)(R_+P) + (E^+ - 1)(R_P) \tag{3.2}$$

We now show how the macroscopic dynamics and the fluctuations around them emerge from the Master equation when the system volume goes to infinity.

3.2 System-Size Expansion to the Linear Fokker–Planck Equation

The Ω-expansion Taylor-expands the Master equation near macroscopic trajectories or stationary solutions in powers of $\Omega^{-1/2}$. The leading-order terms reproduce the macroscopic rate equation, and the next order terms give the Linear Noise Approximation.

To separate the macroscopic dynamics from fluctuations, define

$$n(t) = \Omega x(t) + \sqrt{\Omega}\,\xi(t)$$

Equivalently,

$$\frac{n}{\Omega} = x + \Omega^{-1/2}\xi$$

Let $\Pi(\xi, t)$ denote the probability density of the fluctuation variable ξ. The probability distribution $P_n(t)$ is related to $\Pi(\xi, t)$ through the change of variables,

$$\frac{\partial P_n}{\partial t} = \frac{\partial \Pi}{\partial t} + \frac{\partial \Pi}{\partial \xi}\frac{\partial \xi}{\partial t} = \frac{\partial \Pi}{\partial t} - \sqrt{\Omega}\,\dot{x}\frac{\partial \Pi}{\partial \xi}. \tag{3.3}$$

Since at fixed molecule number

$$\frac{\partial n}{\partial t} = \Omega\dot{x} + \sqrt{\Omega}\frac{\partial \xi}{\partial t} = 0$$

gives

$$\frac{\partial \xi}{\partial t} = -\sqrt{\Omega}\,\dot{x}.$$

Taylor expansion of the mesoscopic transition rates about the macroscopic concentration x:

$$\begin{aligned}\Omega^{-1}R_{\pm}(\Omega x + \sqrt{\Omega}\xi;\Omega) &= r_{\pm}\left(x + \Omega^{-1/2}\xi\right)\\ &= r_{\pm}(x) + \Omega^{-1/2}\xi r'_{\pm}(x) + O(\Omega^{-1}).\end{aligned} \tag{3.4}$$

A change of one molecule corresponds to

$$n \to n \pm 1 \qquad \Longleftrightarrow \qquad \xi \to \xi \pm \Omega^{-1/2}$$

The shift operators can therefore be Taylor-expanded as

$$\begin{aligned}E^{\pm}f(\xi) &= f(\xi \pm \Omega^{-1/2})\\ &= 1 \pm \Omega^{-1/2}\frac{\partial}{\partial \xi} + \frac{1}{2}\Omega^{-1}\frac{\partial^2}{\partial \xi^2} + O(\Omega^{-3/2}).\end{aligned} \tag{3.5}$$

Substituting Eqs. (3.3), (3.4), and (3.5) into the Master equation (3.2) gives

$$\begin{aligned}\frac{\partial p_n}{\partial t} &= \frac{\partial \Pi}{\partial t} - \sqrt{\Omega}\,\dot{x}\frac{\partial \Pi}{\partial \xi} \\ &= \left[-\Omega^{-1/2}\frac{\partial}{\partial \xi} + \frac{1}{2}\Omega^{-1}\frac{\partial^2}{\partial \xi^2} + O(\Omega^{-3/2})\right] \times \Omega\left[r_+(x) + \Omega^{-1/2}\xi r'_+(x) + O(\Omega^{-1})\right]\Pi \\ &\quad + \left[\Omega^{-1/2}\frac{\partial}{\partial \xi} + \frac{1}{2}\Omega^{-1}\frac{\partial^2}{\partial \xi^2} + O(\Omega^{-3/2})\right] \times \Omega\left[r_-(x) + \Omega^{-1/2}\xi r'_-(x) + O(\Omega^{-1})\right]\Pi\end{aligned}$$

The differential operators act on all factors to their right.

Identifying terms of order $\Omega^{1/2}$ gives

$$-\sqrt{\Omega}\,\dot{x}\frac{\partial \Pi}{\partial \xi} = -\sqrt{\Omega}\left[r_+(x) - r_-(x)\right]\frac{\partial \Pi}{\partial \xi}.$$

These terms cancel when

$$\dot{x} = r_+(x) - r_-(x)$$

which is exactly the macroscopic rate equation.

Identifying terms of order Ω^0 gives

$$\frac{\partial \Pi}{\partial t} = -\left[r'_+(x) - r'_-(x)\right]\frac{\partial}{\partial \xi}\left(\xi\Pi\right) + \frac{1}{2}\left[r_+(x) + r_-(x)\right]\frac{\partial^2 \Pi}{\partial \xi^2} \tag{3.6}$$

Define

$$A = \frac{d}{dx}\left[r_+(x) - r_-(x)\right] = r'_+(x) - r'_-(x),$$

$$B^2 = r_+(x) + r_-(x)$$

Equation (3.6) becomes

$$\frac{\partial \Pi}{\partial t} = -\frac{\partial}{\partial \xi}\left(A\xi\Pi\right) + \frac{1}{2}B^2\frac{\partial^2 \Pi}{\partial \xi^2} \tag{3.7}$$

This is the **linear Fokker–Planck equation**. Its drift is linear in the fluctuation ξ, while A and B^2 depend on time through the macroscopic trajectory $x(t)$. Terms of order $\Omega^{-1/2}$ and smaller have been discarded.

3.3 The Stochastic Differential Equation

The linear Fokker–Planck equation (3.7) can equivalently be formulated as the linear Langevin equation as

$$d\xi = A\xi\,dt + B\,dW(t),$$

where $W(t)$ is a standard Wiener process satisfying

$$\langle dW \rangle = 0, \qquad \langle dW^2 \rangle = dt$$

Appendix A. Terms to check after class

1. Stochasticity: What is the different between "stochastic" and "random"?
2. Identical
3. C. V. (coefficient of variation)
4. Propagation of noise
5. Birth – Death process
6. Counting process and shot noise
7. Genome locus
8. Extrinsic vs Intrinsic noise
9. Detailed balance
10. Stationary condition
11. Poisson distribution
12. Chemical Masters Equation (CME)
13. Gillespie Algorithm
14. Partial Differential Equation (PDE)
15. Shift operator
16. van Kampen
17. Omega expansion
18. Linear Noise Approximation (LNA)

References

1. J. Paulsson. Summing up the noise in gene networks. *Nature* **427**, 415–418 (2004). doi:10.1038/nature02257.
2. J. Elf and M. Ehrenberg. Fast Evaluation of Fluctuations in Biochemical Networks With the Linear Noise Approximation. *Genome Research* **13**, 2475–2484 (2003). doi:10.1101/gr.1196503.
3. Michael Elowitz and Justin Bois's course on Biological Circuit Design, Chapter 16-17: https://biocircuits.github.io/